

Quiz

1. Which of the following best describes the relationship between AI, machine learning, and deep learning?

- A. AI is a subset of machine learning, which is a subset of deep learning.
- B. Machine learning is a subset of AI, and deep learning is a subset of machine learning.
- C. Deep learning is a subset of AI, and machine learning is a subset of deep learning.
- D. They are all independent fields with no overlap.

Answer: B

Explanation: AI is the broad goal, machine learning is a technique to achieve AI, and deep learning is a specialized technique within machine learning.

2. What is the primary function of machine learning?

- A. To explicitly program computers to perform specific tasks.
- B. To enable machines to learn from data without explicit programming.
- C. To simulate the human brain in its entirety.
- D. To generate entirely new content like text and images.

Answer: B

Explanation: Machine learning allows machines to learn from data and improve their performance without being specifically programmed for each task.

3. Which of the following is a key characteristic of deep learning?

- A. It relies on explicit programming for every task.
- B. It uses simple algorithms to process data.
- C. It uses neural networks with multiple layers to mimic the human brain.
- D. It is primarily used for outlier detection.

Answer: C

Explanation: Deep learning utilizes artificial neural networks with many layers to analyze data in a way that is inspired by the human brain.

4. What is the main function of generative AI?

- A. Analyzing existing data to find patterns and anomalies.
- B. Predicting future outcomes based on historical data.
- C. Generating new content, such as text, audio, and video.
- D. Optimizing existing systems for improved efficiency.

Answer: C

Explanation: Generative AI focuses on creating new content rather than simply analyzing or predicting from existing data.

5. What are foundation models, like large language models, used for?

- A. To provide a basic framework for traditional programming.
- B. To drive generative AI by creating new content.
- C. To detect outliers in large datasets.
- D. To build expert systems with predefined rules.

Answer: B

Explanation: Foundation models, especially large language models, are the driving force behind generative AI, enabling the creation of diverse content.